

Button

DETECT

INPUT

What does it do?

A button lets someone tell your program to do something the moment they press.

Inside are two metal contacts. Pressing pushes them together so electricity can flow, and letting go springs them apart again.

Your projects can use it to start something, stop something, or count how many times it has been pressed.

Uses



Pedestrian crossing

DETECT
Button is pressed



DECIDE
Someone is waiting



DO
Change the lights



Do not press this

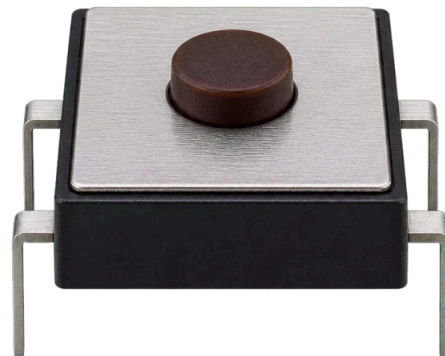
DETECT
Button is pressed



DECIDE
They could not resist



DO
Play a silly sound



Cap

Press here

Legs

Four, joined in pairs

Gap

Straddle the channel

Code

● Raspberry Pi

```
from gpiozero import Button  
  
button = Button(24)  
  
pressed = button.is_pressed
```

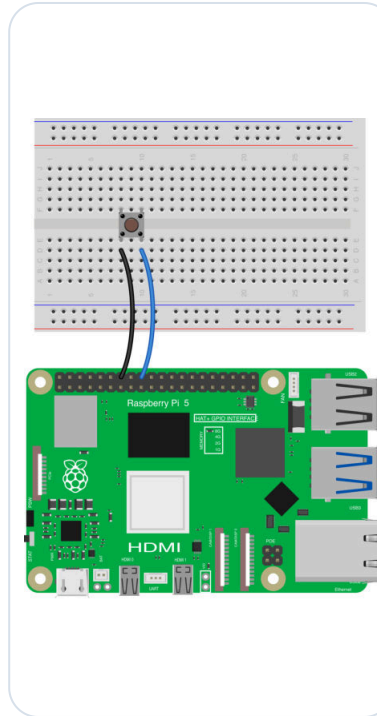
● Raspberry Pi Pico

```
from picozero import Button  
  
button = Button(16)  
  
pressed = button.is_pressed
```

`pressed` is True or False. No resistor is needed — both libraries switch on the pin's own pull-up. Also `button.when_pressed` and `button.when_released`.

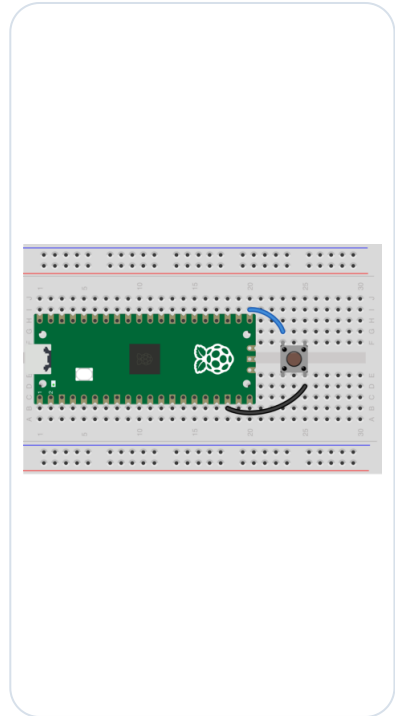
Wiring

RASPBERRY PI 5



GPIO 24 → button → GND

PICO



GP16 → button → GND

Capacitive touch

DETECT

INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

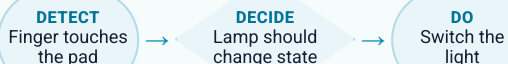
It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

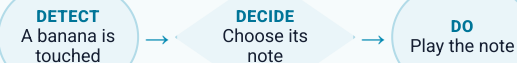
Uses



Touch lamp



Banana piano



Touch pad
The large
metal spiral

SIG
Digital output

**VCC and
GND**
3.3 V power

Code

● Raspberry Pi

```
from gpiozero import DigitalInputDevice

touch = DigitalInputDevice(16, pull_up=False)

touched = touch.is_active
```

● Raspberry Pi Pico

```
from picozero import TouchSensor

touch = TouchSensor(22)

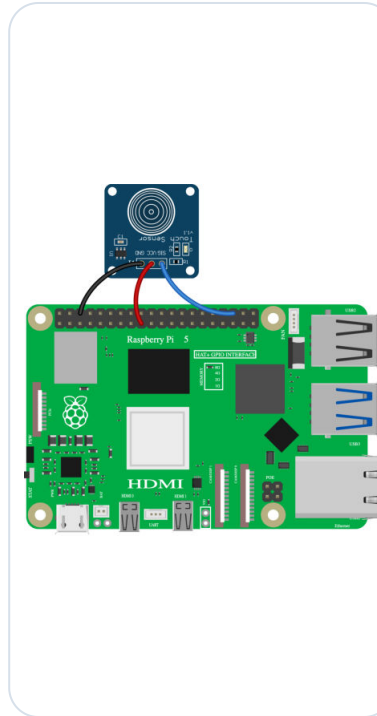
touched = touch.is_touched
```

The TTP223 output goes high while touched.

touched is True or False.

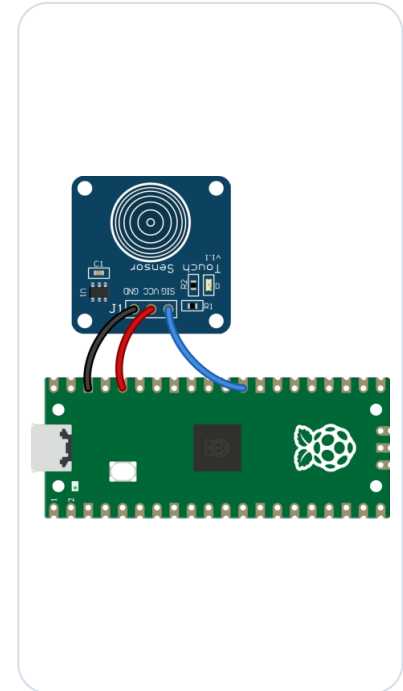
Wiring

RASPBERRY PI 5



GND → GND, **3V3** → VCC, **GPIO 16** → SIG

PICO



GND → GND, **3V3 OUT** → VCC, **GP22** → SIG

LDR

DETECT

LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses



Street light

DETECT

How dark it is



DECIDE

Darker than dusk



DO

Switch the lamp on



Biscuit tin alarm

DETECT

Light inside the tin



DECIDE

The lid is open



DO

Sound the alarm



Window

Light lands here

Track

Zigzag sensor

Legs

Either way round

Code

● Raspberry Pi

```
from gpiozero import LightSensor

ldr = LightSensor(18)

light = ldr.value
```

● Raspberry Pi Pico

```
from picozero import AnalogInputDevice

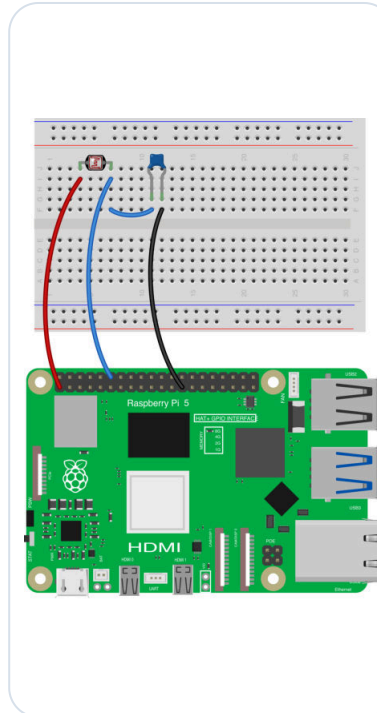
ldr = AnalogInputDevice(26)

light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light. A Pi also gives you `ldr.light_detected` and `ldr.when_light`; a Pico gives you `ldr.voltage`.

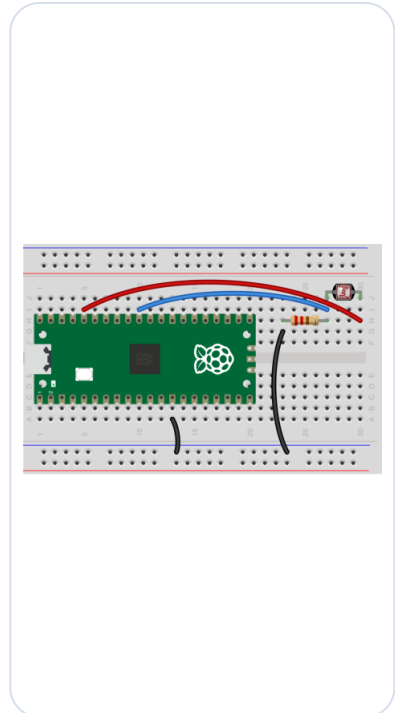
Wiring

RASPBERRY PI 5



3V3 → LDR → GPIO 18 → 1 μ F → GND

PICO



3V3 → LDR → GP26 → 10 k Ω → GND

Line sensor

DETECT

INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

An infrared LED shines at the floor and a detector measures how much light bounces back.

Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses



Line-following robot

DETECT

Read line or floor

DECIDE

Which way to steer

DO

Adjust the motors



Secret floor rune

DETECT

Find a dark mark

DECIDE

Is the rune complete

DO

Reveal a secret



Infrared pair

Faces the floor

Adjuster

Sets the threshold

S

Digital signal

Code

● Raspberry Pi

```
from gpiozero import LineSensor  
  
line = LineSensor(16)  
  
on_line = line.is_active
```

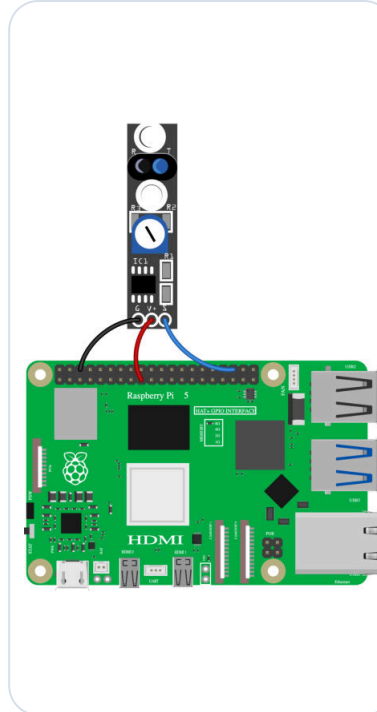
● Raspberry Pi Pico

```
from picozero import DigitalInputDevice  
  
line = DigitalInputDevice(22, pull_up=False)  
  
on_line = line.is_active
```

Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

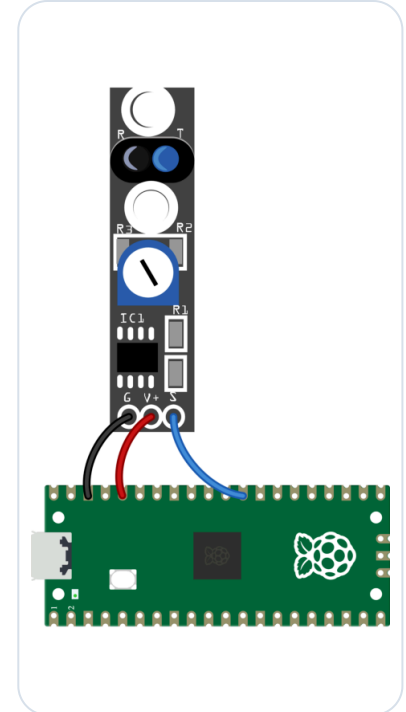
Wiring

RASPBERRY PI 5



GND → GND, **3V3** → V+, **GPIO 16** → S

PICO



GND → GND, **3V3 OUT** → V+, **GP22** → S

Distance sensor

DETECT

DISTANCE

What does it do?

A distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

Uses



Parking sensor

DETECT

Gap to the wall

DECIDE

Closer than
30 cm

DO

Beep faster



Air harp

DETECT

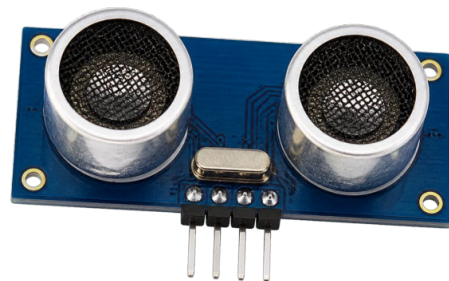
How far your
hand is

DECIDE

Which note
that is

DO

Play the note



VCC

Power,
3V3

Trig

Sends a
click

Echo

Times the
return

GND

Ground

Code

● Raspberry Pi

```
from gpiozero import DistanceSensor

sensor = DistanceSensor(
    echo=18, trigger=17
)

distance = sensor.distance
```

● Raspberry Pi Pico

```
from picozero import DistanceSensor

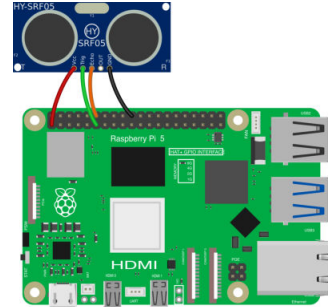
sensor = DistanceSensor(
    echo=15, trigger=13
)

distance = sensor.distance
```

distance is in metres and stops at 1. Pass **max_distance=2** to see further, or read **sensor.value** for 0 to 1. Power it from **3V3**, not 5V.

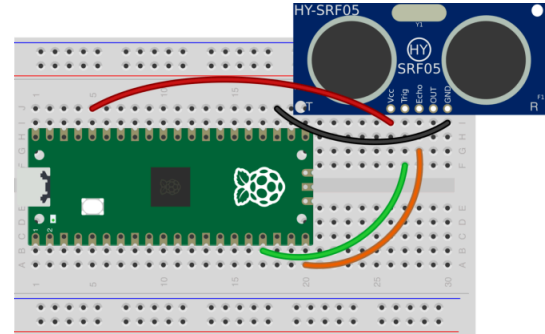
Wiring

RASPBERRY PI 5



Trig **GPIO 17** · Echo **GPIO 18**

PICO



Trig **GP13** · Echo **GP15**

PIR

DETECT

MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses

Burglar alarm

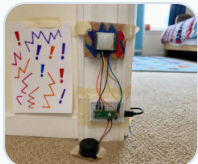


DETECT
Movement in
the room

DECIDE
Nobody
should be in

DO
Raise the
alarm

Sibling alarm



DETECT
Movement at
the door

DECIDE
Someone is
coming

DO
Sound the
warning



Dome
Wide-
angle lens

VCC
Power, 5V

OUT
High on
movement

GND
Ground

Code

● Raspberry Pi

```
from gpiozero import MotionSensor  
  
pir = MotionSensor(4)  
  
moved = pir.motion_detected
```

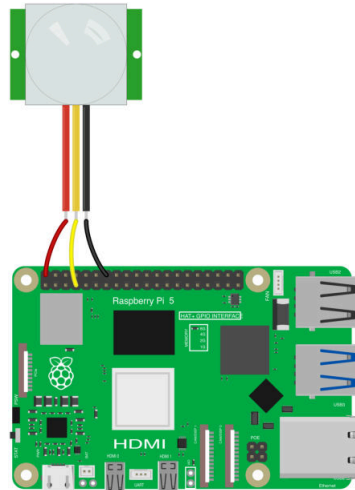
● Raspberry Pi Pico

```
from picozero import MotionSensor  
  
pir = MotionSensor(16)  
  
moved = pir.motion_detected
```

`moved` is True or False. Also `pir.when_motion` and `pir.when_no_motion` to run a function the moment it changes. Most boards carry two dials that set how sensitive it is and how long OUT stays high.

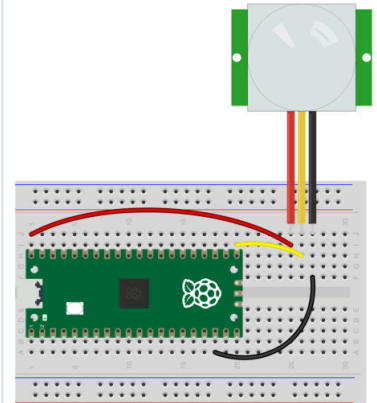
Wiring

RASPBERRY PI 5



5V · OUT GPIO 4 · GND

PICO



VBUS · OUT GP16 · GND

Potentiometer

DETECT

INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses

Sensor sensitivity



DETECT
Knob position

DECIDE
Set trigger level

DO
Adjust sensor response

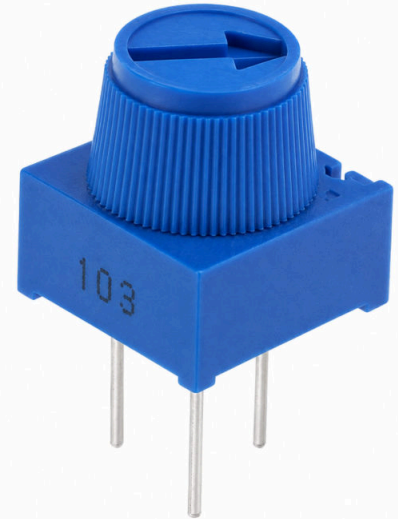
Alien voice tuner



DETECT
Knob position

DECIDE
Choose voice pitch

DO
Rumble or squeak



Knob
Turn to adjust

Outer pair
Pins 1 and 3

Wiper
Pin 2, offset

Code

● Raspberry Pi

```
from gpiozero import MCP3008

pot = MCP3008(0, select_pin=22)

setting = pot.value
```

● Raspberry Pi Pico

```
from picozero import Pot

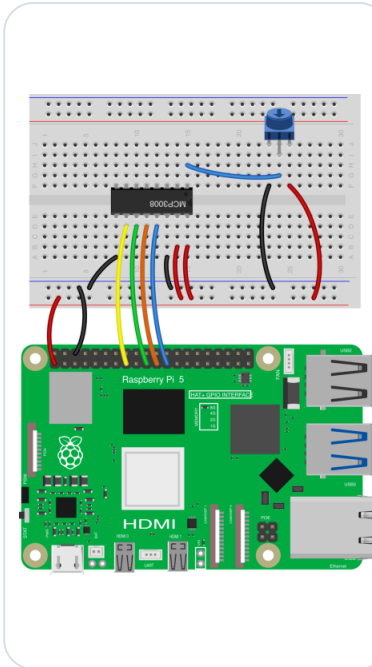
pot = Pot(26)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. The Pi needs the MCP3008 converter shown. Power the trimpot and converter from **3V3**.

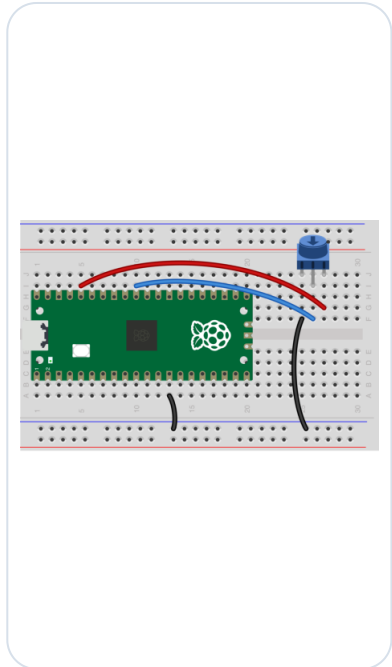
Wiring

PI 5 + MCP3008



Wiper → **CH0**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22**. Chip notch faces right.

PICO



Pin 1 → **GND** · 2 (wiper) → **GP26** · 3 → **3V3**. Each pin has its own breadboard row.

Temperature sensor

DETECT

INPUT

ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Plant propagator

DETECT
Measure the temperature

DECIDE
Too cold or too warm

DO
Control heat or ventilation

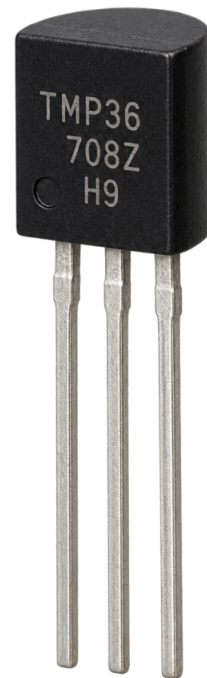


Soup drama meter

DETECT
Measure soup temperature

DECIDE
Safe or lava

DO
Move the drama pointer



Flat face

Use it to identify the pins

+Vs

Left pin

Vout

Centre pin

GND

Right pin

Code

● Raspberry Pi

```
from gpiozero import MCP3008

sensor = MCP3008(channel=0)

temperature = (sensor.voltage - 0.5) * 100
```

● Raspberry Pi Pico

```
from picozero import TemperatureSensor

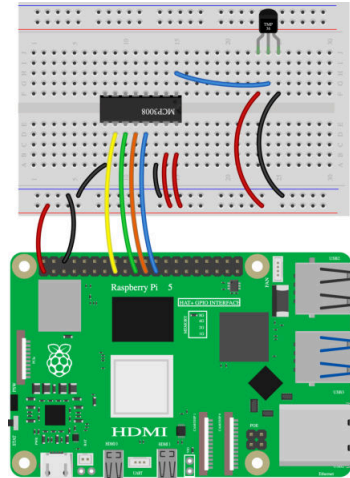
convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(26, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

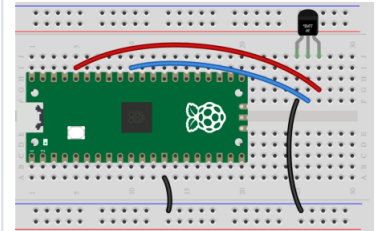
Wiring

RASPBERRY PI 5



TMP36 **Vout** → MCP3008 **CHO**; ADC SPI uses GPIO **22, 10, 9, 11**

PICO



TMP36 **+Vs** → 3V3 OUT, **Vout** → GP26, **GND** → GND

Tilt switch

DETECT

INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

A tiny metal ball rolls inside and touches two contacts, completing the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses



Box lid sensor

DETECT
Lid changes
angle

DECIDE
Box has
opened

DO
Log or sound
an alert



Grumpy pencil case

DETECT
Case is tipped

DECIDE
Pencils are
escaping

DO
Make a
grumpy noise



Metal can
Contains the
rolling ball

Legs
No polarity

Angle
Changes the
contact

Code

● Raspberry Pi

```
from gpiozero import Button

tilt = Button(23)

tilted = tilt.is_pressed
```

● Raspberry Pi Pico

```
from picozero import Switch

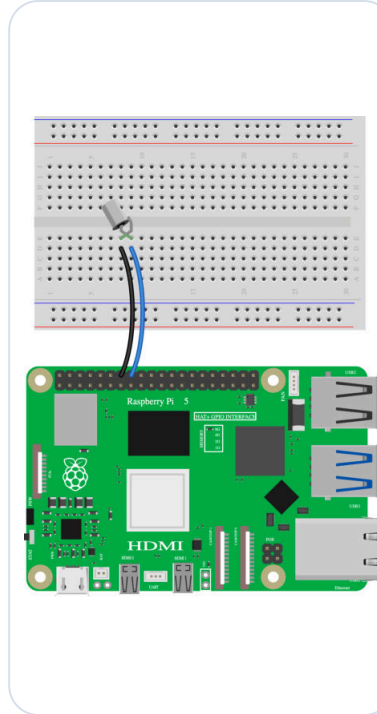
tilt = Switch(16)

tilted = tilt.is_closed
```

The reading may flicker while the ball rolls.
Check it more than once before treating it as a definite tilt.

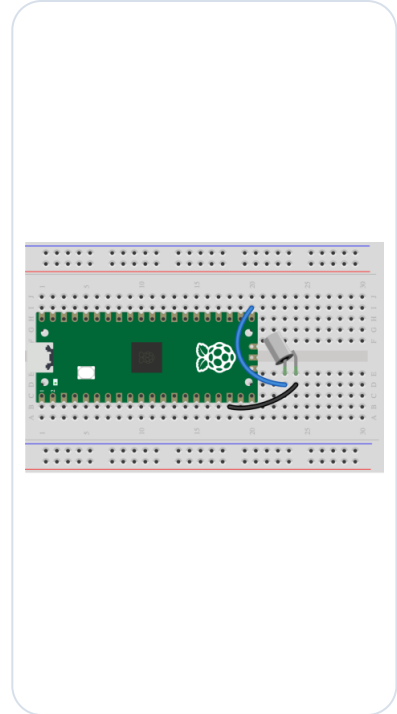
Wiring

RASPBERRY PI 5



GPIO 23 → tilt switch → GND

PICO



GP16 → tilt switch → GND

Toggle switch

DETECT

INPUT

What does it do?

A toggle switch stays in the position you move it to, unlike a push button.

Its centre COM leg connects to one outside leg in each position.

It is useful when a project needs a lasting on/off choice or two modes.

Uses



Mode selector

DETECT

Read the switch position

DECIDE

Choose a mode

DO

Run that behaviour



Serious business lever

DETECT

Lever is thrown

DECIDE

Serious mode selected

DO

Start unnecessary sirens



Lever

Stays where it is moved

COM

Centre pin

L1 and L2

The two choices

Code

● Raspberry Pi

```
from gpiozero import Button  
  
mode = Button(23)  
  
mode_two = mode.is_pressed
```

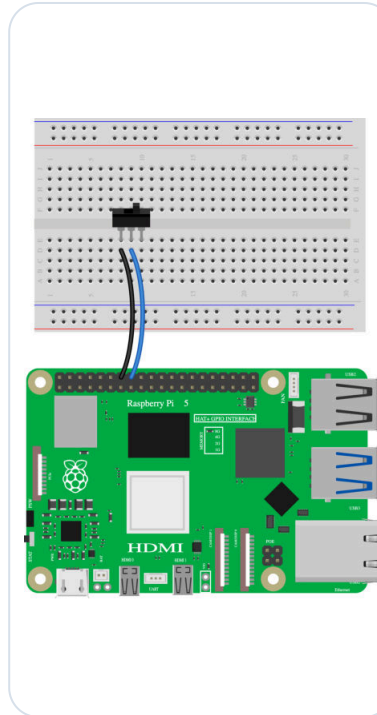
● Raspberry Pi Pico

```
from picozero import Switch  
  
mode = Switch(16)  
  
mode_two = mode.is_closed
```

Connect COM and one outside pin. The unused outside pin can be left unconnected.

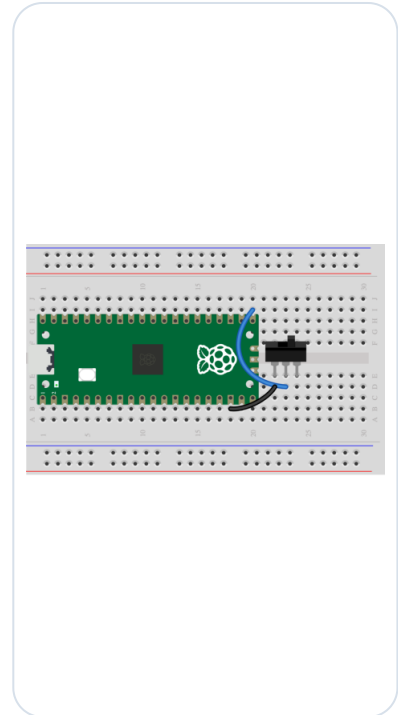
Wiring

RASPBERRY PI 5



GPIO 23 → COM; one outside pin → **GND**

PICO



GP16 → COM; one outside pin → **GND**

LED

DO

LIGHT

What does it do?

An LED is a tiny light that your code can switch on, switch off, or blink.

LED stands for light-emitting diode. It glows when electricity flows the right way through it, so the long leg and the short leg matter.

Your projects can use LEDs to show a warning, celebrate something, or tell people what is happening.

Uses



Car brake light

DETECT

Brake pedal pressed



DECIDE

The car is slowing



DO

Light the red LED



One-eyed monster box

DETECT

Box lid is opened



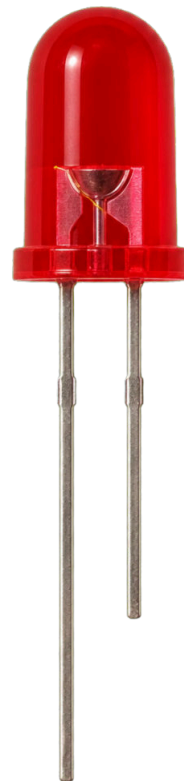
DECIDE

Someone is snooping



DO

Glow the monster eye



Anode

Long leg, to
GPIO

Cathode

Short leg, to
GND

Lens

Glow here

Code

● Raspberry Pi

```
from gpiozero import LED  
  
led = LED(17)  
  
led.on()
```

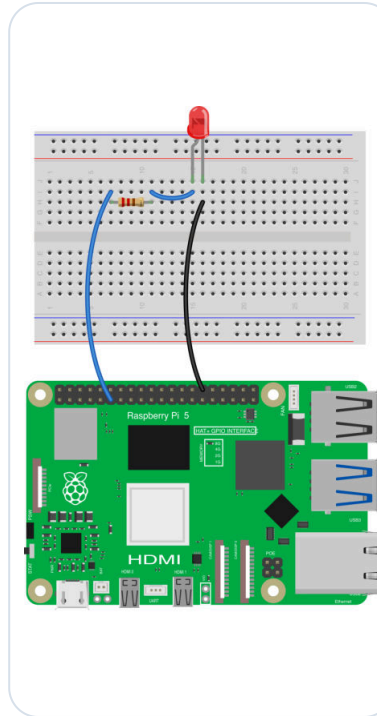
● Raspberry Pi Pico

```
from picozero import LED  
  
led = LED(16)  
  
led.on()
```

Also `led.off()`, `led.toggle()` and `led.blink()`. Give `blink()` an `on_time` and an `off_time` to set the flash speed.

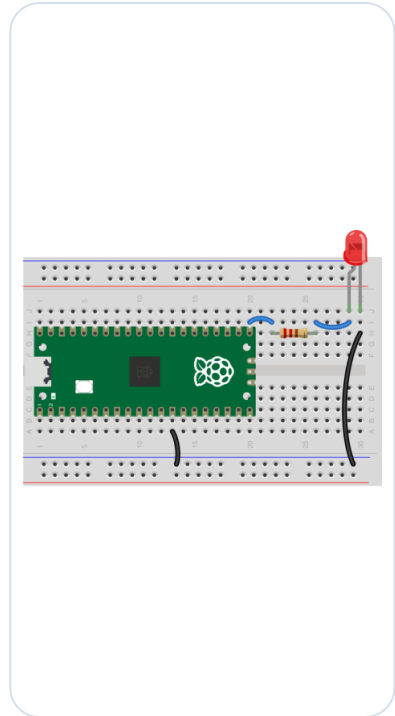
Wiring

RASPBERRY PI 5



GPIO 17 → 220 Ω → LED → GND

PICO



GP16 → 220 Ω → LED → GND

RGB LED

DO

OUTPUT

LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses

Status light



DETECT
Read the
machine state

DECIDE
Choose a
status colour

DO
Mix the LED
colour

Mood crystal



DETECT
Read a
mysterious
input

DECIDE
Pick the
crystal's
mood

DO
Glow that
colour



Clear lens
Contains three
LEDs

Long leg
Common
cathode

R
One 220 ohm
resistor each

Code

● Raspberry Pi

```
from gpiozero import RGBLED
light = RGBLED(red=17, green=27, blue=22)
light.color = (1, 0, 1)
```

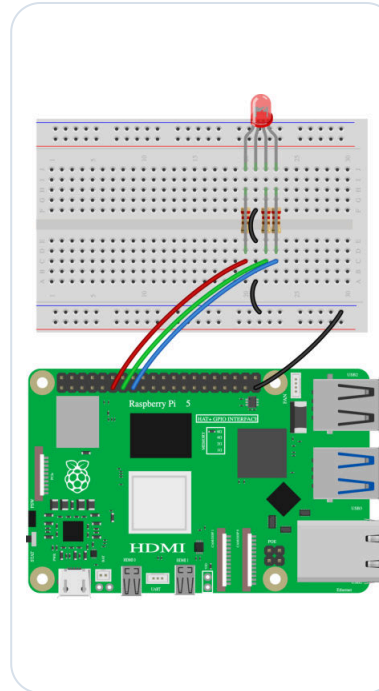
● Raspberry Pi Pico

```
from picozero import RGBLED
light = RGBLED(red=15, green=13, blue=10)
light.color = (255, 0, 255)
```

Pi colour values run from 0 to 1; picozero colour values run from 0 to 255. These examples make magenta.

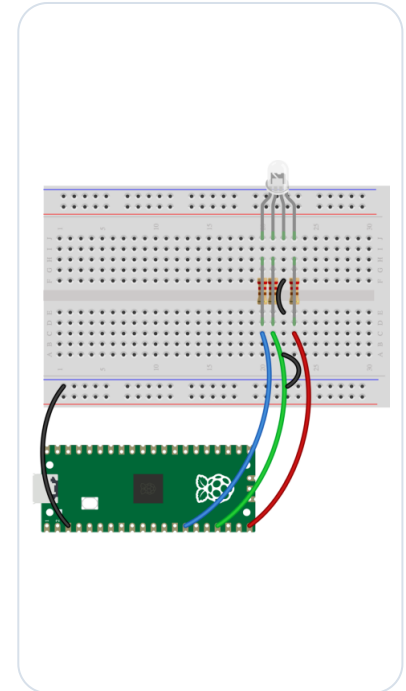
Wiring

RASPBERRY PI 5



R → **GPIO17**, G → **GPIO27**, B → **GPIO22**;
common → **GND**

PICO



R → **GP15**, G → **GP13**, B → **GP10**;
common → **GND**

NeoPixels

DO

OUTPUT

LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses



Progress bar

DETECT
Measure
progress

DECIDE
How many
pixels to fill

DO
Light the strip



Disco shoelace meter

DETECT
Measure
dance energy

DECIDE
Choose
colours and
pattern

DO
Animate the
pixels



Pixel
RGB LED
and
controller

IN
Data from
the board

OUT
Data to
another
pixel

**VDD
and
GND**
3.3 V
power

Code

● Raspberry Pi

```
import board
import neopixel

pixels = neopixel.NeoPixel(board.D18, 1)

pixels[0] = (255, 0, 80)
```

● Raspberry Pi Pico

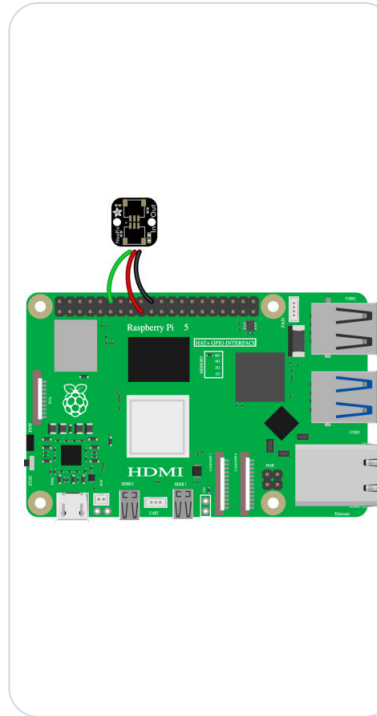
```
from machine import Pin
from neopixel import NeoPixel

pixels = NeoPixel(Pin(28), 1)
pixels[0] = (255, 0, 80)
pixels.write()
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

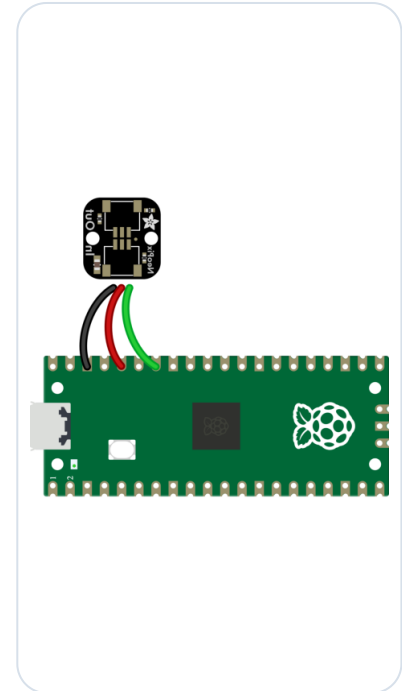
Wiring

RASPBERRY PI 5



GPIO18 → IN, **3V3** → VDD, **GND** → GND

PICO



GP28 → IN, **3V3 OUT** → VDD, **GND** → GND

Buzzer

DO

SOUND

What does it do?

A buzzer makes a sound the moment your code switches it on.

An active buzzer has its tone built in, so it beeps at one fixed pitch. You switch it on and off rather than choosing notes.

Your projects can use it for an alarm, a warning, a countdown, or a fanfare when somebody wins.

Uses



Smoke alarm

DETECT

Smoke in the air



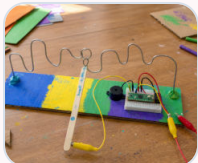
DECIDE

There is a fire



DO

Sound the alarm



Buzz wire game

DETECT

Loop touches wire



DECIDE

A wobble, so out



DO

Buzz loudly



+ leg

Longer, to
GPIO

- leg

Shorter, to GND

Vent

Sound comes
out

Code

● Raspberry Pi

```
from gpiozero import Buzzer  
  
buzzer = Buzzer(12)  
  
buzzer.on()
```

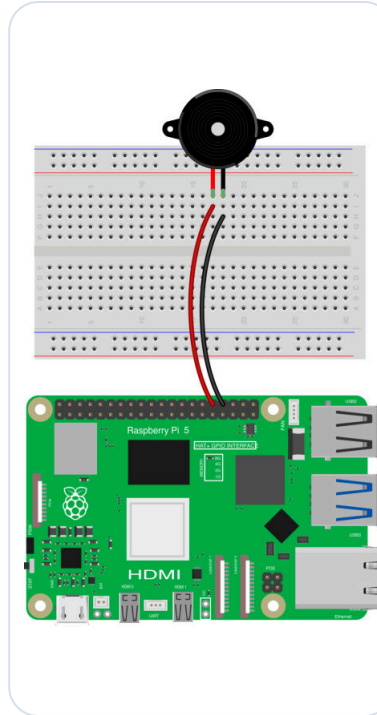
● Raspberry Pi Pico

```
from picozero import Buzzer  
  
buzzer = Buzzer(16)  
  
buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm.

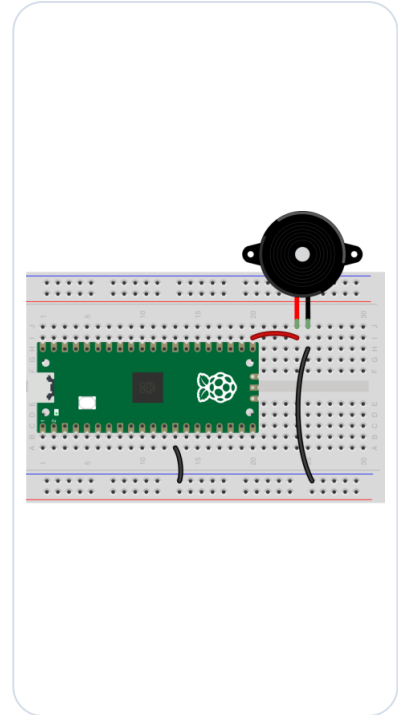
Wiring

RASPBERRY PI 5



GPIO 12 → + buzzer - → GND

PICO



GP16 → + buzzer - → GND

Speaker

DO

OUTPUT

SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses



Door chime

DETECT
Door button is pressed

DECIDE
Which chime to use

DO
Play the notes



Tiny fanfare machine

DETECT
Something mildly impressive happens

DECIDE
It deserves a fanfare

DO
Play triumphant notes



Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Code

● Raspberry Pi

```
from gpiozero import TonalBuzzer  
speaker = TonalBuzzer(17)  
speaker.play("E4")
```

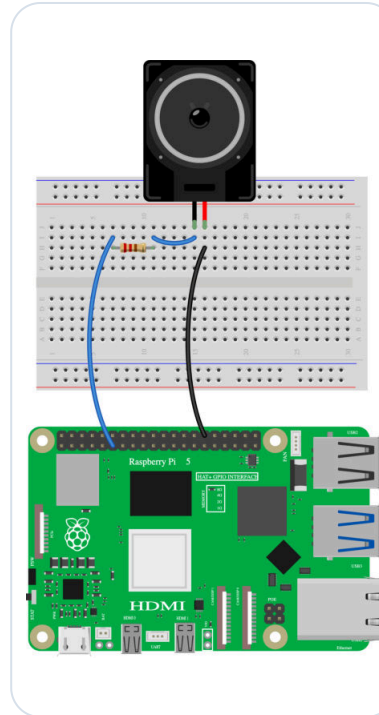
● Raspberry Pi Pico

```
from picozero import Speaker  
speaker = Speaker(16)  
speaker.play("E4", duration=1)
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

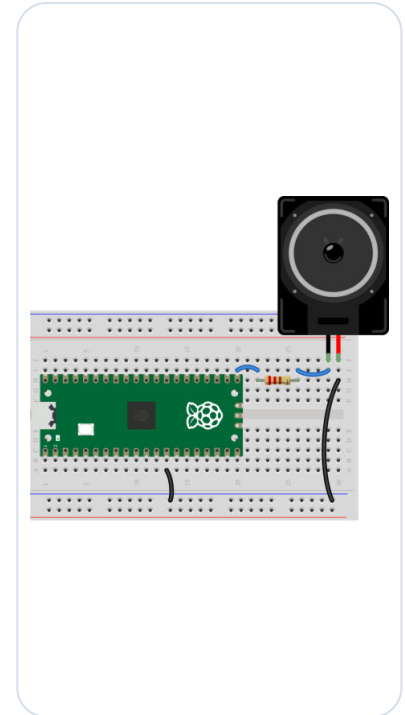
Wiring

RASPBERRY PI 5



GPIO17 → 220 ohms → speaker → **GND**

PICO



GP16 → 220 ohms → speaker → **GND**

Motor driver

DO

OUTPUT

MOVEMENT

What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

The L293D contains H-bridges that can send current through a motor in either direction.

Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses



Robot wheel

DETECT
Read the route
or controls

DECIDE
Direction and
speed

DO
Drive the
wheel

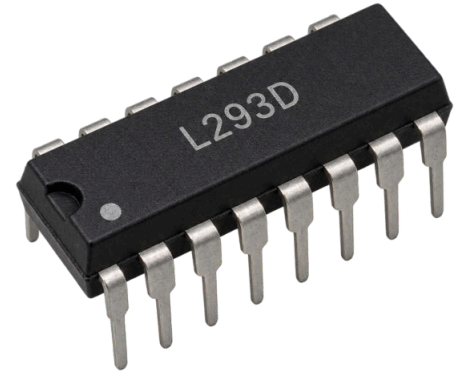


Tiny flag waver

DETECT
Notice a tiny
victory

DECIDE
Is celebration
required

DO
Wave the flag



Notch
Identifies
pin 1

**IN1 and
IN2**
Direction
controls

**OUT1
and
OUT2**
Connect
to the
motor

Enable
Held high
in this
circuit

Code

● Raspberry Pi

```
from gpiozero import Motor
motor = Motor(forward=17, backward=18)
motor.forward()
```

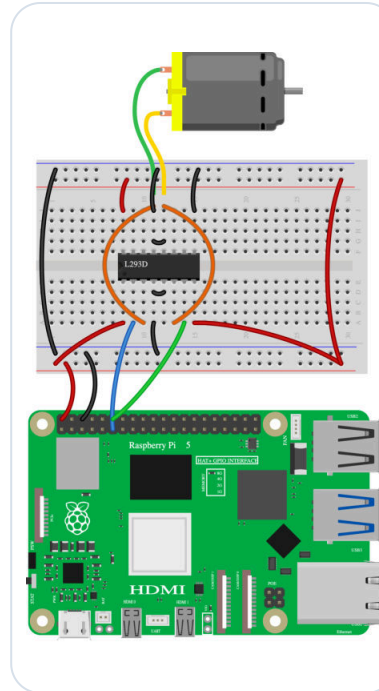
● Raspberry Pi Pico

```
from picozero import Motor
motor = Motor(14, 15)
motor.forward()
```

This card powers one small 5 V motor from the board's 5 V rail. Use a separate motor supply for larger motors, with the grounds joined.

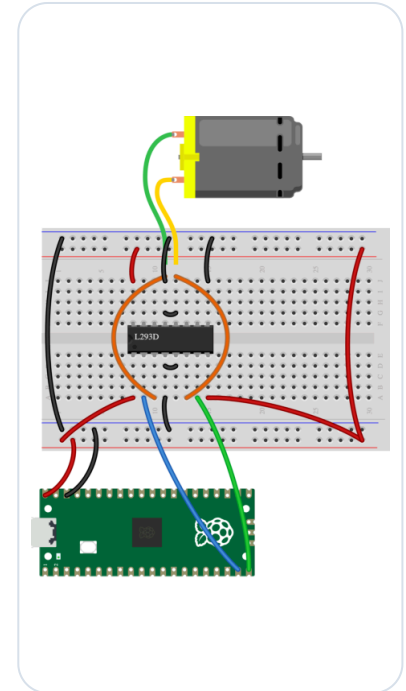
Wiring

RASPBERRY PI 5



GPIO17 → IN1, **GPIO18** → IN2; OUT1 and OUT2 → motor

PICO



GP14 → IN1, **GP15** → IN2; OUT1 and OUT2 → motor

Stepper motor

DO

OUTPUT

MOVEMENT

What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



Pointer dial

DETECT

Read a changing value

DECIDE

Find its dial position

DO

Step the pointer there



Snack lift

DETECT

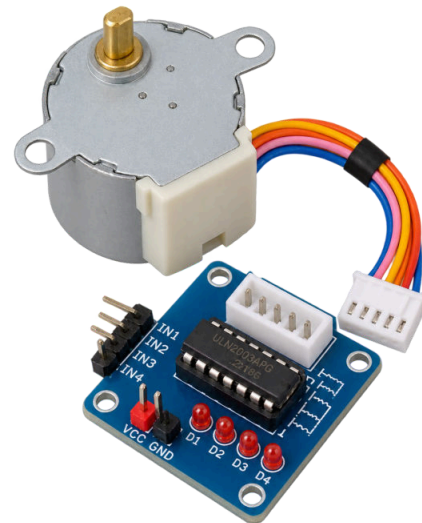
Snack request arrives

DECIDE

Which floor needs it

DO

Step the lift into place



Output shaft

Moves in measured steps

Five-wire plug

Connects to the driver

IN1 to IN4

Coil sequence

5V and GND

Driver power

Code

● Raspberry Pi

```
from gpiozero import OutputDevice
from time import sleep

coils = [OutputDevice(pin) for pin in (22, 27, 17, 4)]
sequence = ((1, 0, 0, 0), (0, 1, 0, 0),
            (0, 0, 1, 0), (0, 0, 0, 1))

for pattern in sequence:
    for coil, value in zip(coils, pattern):
        coil.value = value
    sleep(0.002)
```

● Raspberry Pi Pico

```
from picozero import Stepper

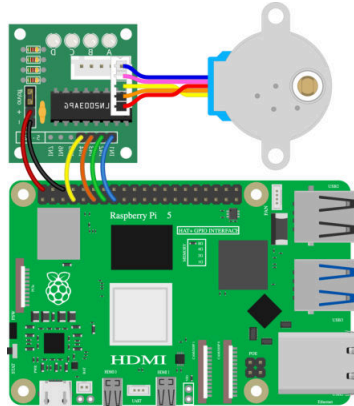
motor = Stepper((16, 17, 18, 19))

motor.step(100)
```

Repeat the Pi sequence to keep moving.
Reverse either board's sequence to turn the other way.

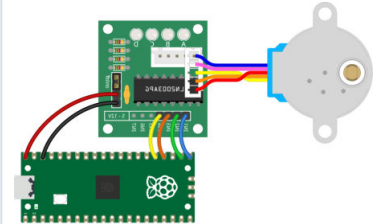
Wiring

RASPBERRY PI 5



IN1-IN4 → GPIO **22, 27, 17, 4**; driver →
5V and GND

PICO



IN1-IN4 → GP **16, 17, 18, 19**; driver →
VBUS and GND